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The present ratio of students to teachers at a certain school is 30 to 1. If the student enrollment were to increase by 50 students and the number of teachers were to increase by 5, the ratio of students to teachers would then be 25 to 1. What is the present number of teachers?

A. 5

B. 8

C. 10

D. 12

E. 15

Algebra Applied Problems

After noting that $\frac{\text{number of students}}{\text{number of teachers}} = \frac{30}{1}$ implies that the number of students is 30 times the number of teachers, this problem can be solved with arithmetic, as shown in the table below, by guessing the present number of teachers, increasing the numbers of students and teachers as specified, then checking to see if the resulting ratio is equal to $\frac{25}{1}$.

Present Number of Teachers	Present Number of Students	Teachers Increased by 5	Students Increased by 50	Resulting Ratio	Equal to $\frac{25}{1}$?
5	150	10	200	$\frac{200}{10} = \frac{20}{1}$	no
8	240	13	290	$\frac{290}{13} = \frac{22\frac{4}{13}}{1}$	no
10	300	15	350	$\frac{350}{15} = \frac{23\frac{1}{3}}{1}$	no
12	360	17	410	$\frac{410}{17} = \frac{24\frac{2}{17}}{1}$	no
15	450	20	500	$\frac{500}{20} = \frac{25}{1}$	yes

Therefore, the present number of teachers is 15.

Alternatively, the problem can be solved using algebra.

The following table summarizes the given information, where s represents the present number of students and t represents the present number of teachers:

	Number of Students	Number of Teachers	$\frac{\text{Students}}{\text{Teachers}}$	Equation After Cross Multiplying
Present	s	t	$\frac{s}{t} = \frac{30}{1}$	$s = 30t$
After Increases	$s + 50$	$t + 5$	$\frac{s+50}{t+5} = \frac{25}{1}$	$s + 50 = 25(t + 5)$

Determine the value of t using $s = 30t$ (Equation 1) and $s + 50 = 25(t + 5) = 25t + 125$ (Equation 2).

$$s + 50 = 25t + 125 \quad \text{Equation 2}$$

$$30t + 50 = 25t + 125 \quad \text{substitution from Equation 1}$$

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$$\begin{aligned} 30t + 50 &= 25t + 125 && \text{substitution from Equation 1} \\ 5t + 50 &= 125 && \text{subtract } 25t \text{ from both sides} \\ 5t &= 75 && \text{subtract 50 from both sides} \\ t &= 15 && \text{divide both sides by 5} \end{aligned}$$

Therefore, the present number of teachers is 15.

The correct answer is E.

•

What is the smallest integer n for which $25^n > 5^{12}$?

A. 6

B. 7

C. 8

D. 9

E. 10

Arithmetic Operations with Rational Numbers

Because $5^2 = 25$, a common base is 5. Rewrite the left side with 5 as a base: $25^n = (5^2)^n = 5^{2n}$. It follows that the desired integer is the least integer n for which $5^{2n} > 5^{12}$. This will be the least integer n for which $2n > 12$, or the least integer n for which $n > 6$, which is 7.

The correct answer is B.

•

Sixty percent of the members of a study group are women, and 45 percent of those women are lawyers. If one member of the study group is to be selected at random, what is the probability that the member selected is a woman lawyer?

A. 0.10

B. 0.15

C. 0.27

D. 0.33

E. 0.45

Arithmetic Probability

For simplicity, suppose there are 100 members in the study group. Since 60 percent of the members are women, there are 60 women in the group. Also, 45 percent of the women are lawyers so there are 0.45(60) = 27 women lawyers in the study group. Therefore, the probability of selecting a woman lawyer is $\frac{27}{100} = 0.27$.

The correct answer is C.

•

Each year for 4 years, a farmer increased the number of trees in a certain orchard by $\frac{1}{4}$ of the number of trees in the orchard the preceding year. If all of the trees thrived and there were 6,250 trees in the orchard at the end of the 4-year period, how many trees were in the orchard at the beginning of the 4-year period?

A. 1,250

B. 1,563

C. 2,250

D. 2,560

E. 2,752

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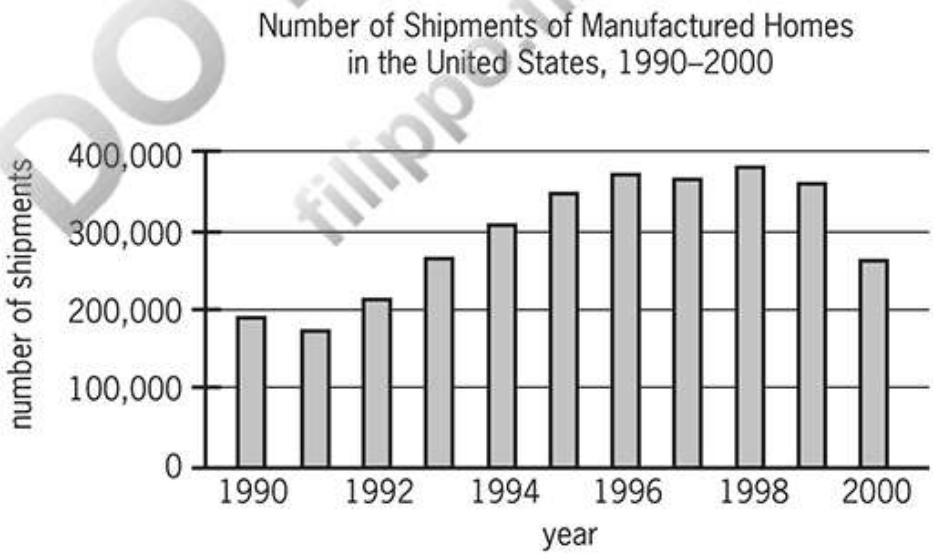
E. 2,732

Arithmetic Operations on Rational Numbers

Let N be the number of trees in the orchard at the beginning of the 4-year period. Since the number of trees increases by a factor of $\frac{1}{4}$ each year, the number of trees after the first year is $N + \frac{1}{4}N = \frac{5}{4}N$. Note that this shows algebraically that an increase by a factor of $\frac{1}{4}$ corresponds to multiplication by $\frac{5}{4}$. Therefore, the number of trees after the second year is $\frac{5}{4}$ times $\frac{5}{4}N$, which equals $\left(\frac{5}{4}\right)^2 N$. Similarly, the number of trees after the third year is $\left(\frac{5}{4}\right)^3 N$ and the number of trees after the fourth year is $\left(\frac{5}{4}\right)^4 N$. Since the number of trees after the fourth year is 6,250, it follows that $\left(\frac{5}{4}\right)^4 N = 6,250$. Next, solve for N:

$\left(\frac{5}{4}\right)^4 N$	=	6, 250	given
$\left(\frac{5^4}{4^4}\right) N$	=	6, 250	property of exponents
N	=	$6,250 \div \frac{5^4}{4^4}$	divide both sides by $\frac{5^4}{4^4}$
N	=	$6,250 \times \frac{4^4}{5^4}$	invert and multiply
N	=	$2 \times 5^5 \times \frac{4^4}{5^4}$	factor 6,250
N	=	$2 \times 5 \times 4^4$	cancel common factors
N	=	2,560	multiply

The correct answer is D.



•

According to the chart shown, which of the following is closest to the median annual number of shipments of manufactured homes in the United States for the years from 1990 to 2000, inclusive?

A. 250,000

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A. 250,000

B. 280,000

C. 310,000

D. 325,000

E. 340,000

Arithmetic Interpretation of Graphs and Tables; Statistics

From the chart, the approximate numbers of shipments are as follows:

Year	Number of Shipments
1990	190,000
1991	180,000
1992	210,000
1993	270,000
1994	310,000
1995	350,000
1996	380,000
1997	370,000
1998	390,000
1999	360,000
2000	270,000

Since there are 11 entries in the table and 11 is an odd number, the median of the numbers of shipments is the 6th entry when the numbers of shipments are arranged in order from least to greatest. In order, from least to greatest, the first 6 entries are:

Number of Shipments
180,000
190,000
210,000
270,000
270,000
310,000

The 6th entry is 310,000.

The correct answer is C.

•

For the positive integers a, b, and k, $a^k \| b$ means that a^k is a divisor of b, but a^{k+1} is not a divisor of b. If k is a positive integer and $2^k \| 72$, then k is equal to

A. 2

B. 3

C. 4

D. 8

E. 18

Arithmetic Property of Numbers

Since $72 = (2^3)(3^2)$, it follows that 2^3 is a divisor of 72 and 2^4 is not a divisor of 72. Therefore, $2^3 \| 72$, and hence

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Since $72 = (2^3)(3^2)$, it follows that 2^3 is a divisor of 72 and 2^4 is not a divisor of 72. Therefore, $2^3 \| 72$, and hence $k = 3$.

The correct answer is B.

•

A certain characteristic in a large population has a distribution that is symmetric about the mean m. If 68 percent of the distribution lies within one standard deviation d of the mean, what percent of the distribution is less than $m + d$?

A. 16%

B. 32%

C. 48%

D. 84%

E. 92%

Arithmetic Statistics

Since 68% lies between $m - d$ and $m + d$, a total of (100 − 68)% = 32% lies to the left of $m - d$ and to the right of $m + d$. Because the distribution is symmetric about m, half of the 32% lies to the right of $m + d$. Therefore, 16% lies to the right of $m + d$, and hence (100 − 16)% = 84% lies to the left of $m + d$.

The correct answer is D.

•

Four extra-large sandwiches of exactly the same size were ordered for m students, where $m > 4$. Three of the sandwiches were evenly divided among the students. Since 4 students did not want any of the fourth sandwich, it was evenly divided among the remaining students. If Carol ate one piece from each of the four sandwiches, the amount of sandwich that she ate would be what fraction of a whole extra-large sandwich?

A. $\frac{m+4}{m(m-4)}$

B. $\frac{2m-4}{m(m-4)}$

C. $\frac{4m-4}{m(m-4)}$

D. $\frac{4m-8}{m(m-4)}$

E. $\frac{4m-12}{m(m-4)}$

Algebra Applied Problems

Since each of 3 of the sandwiches was evenly divided among m students, each piece was $\frac{1}{m}$ of a sandwich. Since the fourth sandwich was evenly divided among $m - 4$ students, each piece was $\frac{1}{m-4}$ of the fourth sandwich. Carol ate 1 piece from each of the four sandwiches, so she ate a total of

$$(3)\frac{1}{m} + \frac{1}{m-4} = \frac{3(m-4)+m}{m(m-4)} = \frac{4m-12}{m(m-4)}$$

The correct answer is E.

•

Which of the following equations has $1 + \sqrt{2}$ as one of its roots?

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A. $x^2 + 2x - 1 = 0$

B. $x^2 - 2x + 1 = 0$

C. $x^2 + 2x + 1 = 0$

D. $x^2 - 2x - 1 = 0$

E. $x^2 - x - 1 = 0$

Algebra Second-Degree Equations

This problem can be solved by working backward to construct a quadratic equation with $1 + \sqrt{2}$ as a root that does not involve radicals.

$x = 1 + \sqrt{2}$	set x to the desired value
$x - 1 = \sqrt{2}$	subtract 1 from both sides
$(x-1)^2 = 2$	square both sides
$x^2 - 2x + 1 = 2$	expand the left side
$x^2 - 2x - 1 = 0$	subtract 2 from both sides

The correct answer is D.

•

In Country C, the unemployment rate among construction workers dropped from 16 percent on September 1, 1992, to 9 percent on September 1, 1996. If the number of construction workers was 20 percent greater on September 1, 1996, than on September 1, 1992, what was the approximate percent change in the number of unemployed construction workers over this period?

A. 50% decrease

B. 30% decrease

C. 15% decrease

D. 30% increase

E. 55% increase

Arithmetic Percents

Letting N represent the number of construction workers on September 1, 1992, the information given in the problem is as follows:

1. On September 1, 1992, the unemployment rate among construction workers was 16 percent, so the **number** of unemployed construction workers on September 1, 1992, was $0.16N$.
2. On September 1, 1996, the **number** of construction workers was 20 percent greater than on September 1, 1992, so the **number** of construction workers on September 1, 1996, was $1.2N$.
3. On September 1, 1996, the unemployment rate among construction workers was 9 percent, so the **number** of unemployed construction workers on September 1, 1996, was $0.09(1.2N)$.

As a fraction, the percent change in the **number** of unemployed construction workers from September 1, 1992, to September 1, 1996, is given by:

$\frac{\text{number in 1996} - \text{number in 1992}}{\text{number in 1992}} = \frac{0.09(1.2N) - 0.16N}{0.16N}$	substitution of expressions from (1) and (3)
$= \frac{0.09(1.2) - 0.16}{0.16}$	divide numerator and denominator by N
$= \frac{9(12) - 160}{160}$	multiply numerator and denominator by 1,000 for ease of calculation
$= \frac{108 - 160}{}$	multiply

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$$= \frac{}{160} \qquad \text{multiply}$$

$$= -\frac{52}{160} \qquad \text{subtract}$$

$$= -\frac{13}{40} \qquad \text{divide numerator and denominator by 4}$$

$$\approx -\frac{13}{39} \qquad 39 \approx 40$$

$$\approx -\frac{1}{3} \qquad \text{divide numerator and denominator by 13}$$

Then, as a percent, $-\frac{1}{3}$ represents a decrease of about 30%.

The correct answer is B.

•

In a box of 12 pens, a total of 3 are defective. If a customer buys 2 pens selected at random from the box, what is the probability that neither pen will be defective?

A. $\frac{1}{6}$

B. $\frac{2}{9}$

C. $\frac{6}{11}$

D. $\frac{9}{16}$

E. $\frac{3}{4}$

Arithmetic Probability

The probability that the first pen selected is not defective is $\frac{9}{12} = \frac{3}{4}$ (9 pens are not defective out of a total of 12 pens). Assuming the first pen selected is not defective, the probability that the second pen is not defective is $\frac{8}{11}$ (this time 8 pens are not defective out of a total of 11 pens). Therefore, using the multiplication rule for dependent events, the probability that both pens are not defective is $\left(\frac{3}{4}\right)\left(\frac{8}{11}\right) = \frac{6}{11}$.

Alternatively, by a direct count using the number of combinations of n objects taken k at a time (see Section 3.4.3, "*Counting Methods,*" in the Math Review chapter), there are $\binom{9}{2} = \frac{(9)(8)}{2} = 36$ ways to select 2 non-defective pens from the 9 non-defective pens, and there are $\binom{12}{2} = \frac{(12)(11)}{2} = 66$ ways to select 2 pens from the 12 pens. Therefore, the probability that the 2 pens selected are not defective is

$$\frac{\text{number of ways to select 2 non-defective pens}}{\text{number of ways to select 2 pens}} = \frac{36}{66} = \frac{6}{11}.$$

The correct answer is C.

•

At a certain fruit stand, the price of each apple is 40 cents and the price of each orange is 60 cents. Mary selects a total of 10 apples and oranges from the fruit stand, and the average (arithmetic mean) price of the 10 pieces of fruit is 56 cents. How many oranges must Mary put back so that the average price of the pieces of fruit that she keeps is 52

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56 cents. How many oranges must Mary put back so that the average price of the pieces of fruit that she keeps is 52 cents?

A. 1

B. 2

C. 3

D. 4

E. 5

Algebra Statistics

This problem can be solved using arithmetic or algebra.

With algebra

Let x be the number of oranges Mary selected originally. It follows that the number of apples Mary selected is $10 - x$ since the number of pieces of fruit she selected originally was 10. The average price of the 10 pieces of fruit is 56 cents so,

$$\begin{aligned}
\frac{40(10-x)+60x}{10} &= 56 && \text{given}\\
40(10-x)+60x &= 560 && \text{multiply both sides by 10}\\
400-40x+60x &= 560 && \text{distributive property}\\
400+20x &= 560 && \text{combine like terms}\\
20x &= 160 && \text{subtract 400 from each term}\\
x &= 8 && \text{divide both sides by 20}
\end{aligned}$$

Thus, Mary originally selected 8 oranges and 2 apples.

If y represents the number of oranges that Mary must put back, then $8 - y$ is the number of oranges she keeps. So, with the 2 apples, the total number of pieces of fruit she keeps is $2 + (8 - y)$. The average price of these $2 + (8 - y)$ pieces of fruit is 52 cents so,

$$\begin{aligned}
\frac{40(2)+60(8-y)}{2+(8-y)} &= 52 && \text{given}\\
40(2)+60(8-y) &= 52[2+(8-y)] && \text{multiply both sides by } 2+(8-y)\\
40(2)+60(8-y) &= 52(10-y) && \text{combine like terms}\\
80+480-60y &= 520-52y && \text{distributive property}\\
560-60y &= 520-52y && \text{combine like terms}\\
560 &= 520+8y && \text{add } 60y \text{ to both sides}\\
40 &= 8y && \text{subtract 520 from both sides}\\
5 &= y && \text{divide both sides by 8}
\end{aligned}$$

Therefore, Mary must put back 5 oranges so that the average price of the 2 apples and 3 oranges she keeps is $\frac{40(2)+60(3)}{5} = \frac{260}{5} = 52$ cents.

With arithmetic

In order to know how many oranges Mary should put back to decrease the average price of the pieces of fruit she is buying to 52 cents, the number of oranges she originally selected needs to be determined. Consider the following table:

Number of apples	Number of oranges	Average price (cents)	Average = 56 cents?
1	9	$\frac{1(40)+9(60)}{10} = \frac{580}{10} = 58$	no

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		$\frac{}{10} = \frac{}{10}$	
2	8	$\frac{2(40)+8(60)}{10} = \frac{560}{10} = 56$	**yes**

Thus, Mary originally selected 2 apples and 8 oranges.

Next, determine how many oranges Mary must put back so that the average price of the pieces of fruit she keeps is 52 cents, consider the following table:

Number of apples	Number of oranges put back	Number of oranges kept	Average price (cents)	Average = 52 cents?
2	1	7	$\frac{2(40)+7(60)}{9} = \frac{500}{9} = 55\frac{5}{9}$	no
2	2	6	$\frac{2(40)+6(60)}{8} = \frac{440}{8} = 55$	no
2	3	5	$\frac{2(40)+5(60)}{7} = \frac{380}{7} = 54\frac{2}{7}$	no
2	4	4	$\frac{2(40)+4(60)}{6} = \frac{320}{6} = 53\frac{1}{3}$	no
2	5	3	$\frac{2(40)+3(60)}{5} = \frac{260}{5} = 52$	**yes**

Therefore, Mary must put back 5 oranges.

Note: It is not necessary to perform all of the calculations shown here. It should be obvious, for example, that $\frac{500}{9}$, $\frac{380}{7}$, and $\frac{320}{6}$ are not whole numbers and therefore cannot equal 52. Nor is it necessary to increment the number of oranges she must put back by 1. For this problem, a good guess might be 4, thinking she might need to put back half the oranges she selected originally. Since putting back 4 oranges doesn't bring the average price down enough, the only other choice is 5, which is the correct answer.

Note that the steps are explained in great detail here. Having good algebraic manipulation skills would make most of the steps fairly automatic.

The correct answer is E.

•

A pharmaceutical company received \$3 million in royalties on the first \$20 million in sales of the generic equivalent of one of its products and then \$9 million in royalties on the next \$108 million in sales. By approximately what percent did the ratio of royalties to sales decrease from the first \$20 million in sales to the next \$108 million in sales?

A. 8%

B. 15%

C. 45%

D. 52%

E. 56%

Arithmetic Percents

The ratio of royalties to sales for the first \$20 million in sales is $\frac{3}{20}$, and the ratio of royalties to sales for the next \$108 million in sales is $\frac{9}{108} = \frac{1}{12}$. Now calculate the percent decrease from $\frac{3}{20}$ to $\frac{1}{12}$.

$$\frac{\frac{1}{12}}{\quad} \cdot 100 = \left(\frac{1}{\quad} - \frac{3}{\quad}\right) \cdot \frac{20}{\quad} \cdot 100$$

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$$\frac{12}{-} \times 100 = \left(\frac{1}{12} - \frac{3}{20}\right) \times \frac{20}{3} \times 100$$
$$= \left(\frac{1 \times 20}{12 \times 3} - \frac{3 \times 20}{20 \times 3}\right) \times 100$$
$$= \left(\frac{5}{9} - 1\right) \times 100$$
$$= -\frac{4}{9} \times 100$$
$$\approx -0.44 \times 100$$
$$\approx 45\% \text{ decrease}$$

An alternative way to calculate the percent decrease from $\frac{3}{20}$ to $\frac{1}{12}$ is to scale the values up until they are integers, which allows for a simpler calculation. Multiplying each of these values by

60 gives $60 \times \frac{3}{20} = 9$ and $60 \times \frac{1}{12} = 5$, respectively, so the problem reduces to calculating the percent decrease from 9 to 5. Therefore, $\frac{5-9}{9} \times 100 = -\frac{4}{9} \times 100 \approx -0.44 \times 100 \approx 45\%$ decrease.

The correct answer is C.

Times at Which the Door Opened from 8:00 to 10:00

8:00	8:06	8:30	9:05
8:03	8:10	8:31	9:11
8:04	8:18	8:54	9:29
8:04	8:19	8:57	9:31

•

The light in a restroom operates with a 15-minute timer that is reset every time the door opens as a person goes in or out of the room. Thus, after someone enters or exits the room, the light remains on for only 15 minutes unless the door opens again and resets the timer for another 15 minutes. If the times listed above are the times at which the door opened from 8:00 to 10:00, approximately how many minutes during this two-hour period was the light off?

A. 10

B. 25

C. 35

D. 40

E. 70

Arithmetic Operations with Integers

In the table given above, read down the columns looking for pairs of consecutive times that are more than 15 minutes apart. The following table shows all such pairs:

Consecutive times more than 15 minutes apart	Minutes between times door opens	Minutes that light is on	Minutes that light is off
8:31 and 8:54	54 – 31 = 23	15	23 – 15 = 8
9:11 and 9:29	29 – 11 = 18	15	18 – 15 = 3
9:31 and 10:00	60 – 31 = 29	15	29 – 15 = 14
Total time light is off			25

The correct answer is B.

•

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If p is the product of the integers from 1 to 30, inclusive, what is the greatest integer k for which 3^k is a factor of p?

A. 10

B. 12

C. 14

D. 16

E. 18

Arithmetic Properties of Numbers

The table below shows the numbers from 1 to 30, inclusive, that have at least one factor of 3 and how many factors of 3 each has.

Multiples of 3 Between 1 and 30	Number of Factors of 3
3	1
6 = 2 × 3	1
9 = 3 × 3	2
12 = 2 × 2× 3	1
15 = 3× 5	1
18 = 2 × 3 × 3	2
21 = 3 × 7	1
24 = 2 × 2× 2× 3	1
27 = 3 × 3 × 3	3
30 = 2 × 3 × 5	1

The sum of the numbers in the right column is 14. Therefore, 3^{14} is the greatest power of 3 that is a factor of the product of the first 30 positive integers.

The correct answer is C.

•

If $n = 3^8 - 2^8$, which of the following is NOT a factor of n?

A. 97

B. 65

C. 35

D. 13

E. 5

Arithmetic Properties of Numbers

Since $3^8 - 2^8$ is the difference of the perfect squares $(3^4)^2$ and $(2^4)^2$, then $3^8 - 2^8 = (3^4+2^4)(3^4 - 2^4)$. But $3^4 - 2^4$ is also the difference of the perfect squares $(3^2)^2$ and $(2^2)^2$ so $3^4 - 2^4 = (3^2+2^2)(3^2 - 2^2)$ and therefore $3^8 - 2^8 = (3^4+2^4)(3^2 + 2^2)(3^2 - 2^2)$. It follows that $3^8 - 2^8$ can be factored as $(81 + 16)(9 + 4)(9 - 4) = (97)(13)(5)$. Therefore, 7 is not a factor of $3^8 - 2^8$, and hence $35 = 5 \times 7$ is not a factor of $3^8 - 2^8$. It is easy to see that each of 97, 13, and 5 is a factor of $3^8 - 2^8$, and so is 65, since $65 = 5 \times 13$, although this additional analysis is not needed to arrive at the correct answer.

The correct answer is C.

•

Club X has more than 10 but fewer than 40 members. Sometimes the members sit at tables with 3 members at one table and 4 members at each of the other tables, and sometimes they sit at tables with 3 members at one table and 5 members at each of the other tables. If they sit at tables with 6 members at each table except one and fewer than 6

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members at that one table, how many members will be at the table that has fewer than 6 members?

A. 1

B. 2

C. 3

D. 4

E. 5

Arithmetic Properties of Numbers

Let n be the number of members that Club X has. Since the members can be equally divided into groups of 4 each with 3 left over, and the members can be equally divided into groups of 5 each with 3 left over, it follows that $n-3$ is divisible by both 4 and 5. Therefore, $n-3$ must be a multiple of (4)(5) = 20. Also, because the only multiple of 20 that is greater than 10 and less than 40 is 20, it follows that $n-3=20$, or $n=23$. Finally, when these 23 members are divided into the greatest number of groups of 6 each, there will be 5 members left over, since 23 = (3)(6) + 5.

The correct answer is E.

•

In order to complete a reading assignment on time, Terry planned to read 90 pages per day. However, she read only 75 pages per day at first, leaving 690 pages to be read during the last 6 days before the assignment was to be completed. How many days in all did Terry have to complete the assignment on time?

A. 15

B. 16

C. 25

D. 40

E. 46

Algebra Applied Problems

Let n be the number of days that Terry read at the slower rate of 75 pages per day. Then $75n$ is the number of pages Terry read at this slower rate, and $75n+690$ is the total number of pages Terry needs to read. Also, $n+6$ is the total number of days that Terry will spend on the reading assignment. The requirement that Terry average 90 pages per day is equivalent to $\frac{75n+690}{n+6}=90$. Then

$$\begin{aligned}\frac{75n+690}{n+6}&=90\\75n+690&=90n+540\\150&=15n\\10&=n\end{aligned}$$

Therefore, the total number of days that Terry has to complete the assignment on time is $n+6=10+6=16$.

The correct answer is B.

•

If $s>0$ and $\sqrt{\frac{r}{s}}=s$, what is r in terms of s?

A. $\frac{1}{s}$

B. $\sqrt{s}$

C. $s\sqrt{s}$

D. s^3

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E. $s^3 - s$

Algebra Equations

Solve the equation for r as follows:

$$\sqrt{\frac{r}{s}} = s$$

$$\frac{r}{s} = s^2 \quad \text{square both sides of the equation}$$

$$r = s^3 \quad \text{multiply both sides by } s$$

The correct answer is D.

•

If $3 < x < 100$, for how many values of x is $\frac{x}{3}$ the square of a prime number?

A. Two

B. Three

C. Four

D. Five

E. Nine

Arithmetic Properties of Numbers

If $\frac{x}{3}$ is the square of a prime number, then possible values of $\frac{x}{3}$ are $2^2, 3^2, 5^2, 7^2, \ldots$ Therefore, possible values of x are $3 \times 2^2 = 12$, $3 \times 3^2 = 27$, $3 \times 5^2 = 75$, $3 \times 7^2 = 147$, …. Since only three of these values, namely 12, 27, and 75, are between 3 and 100, there are three values of x such that $\frac{x}{3}$ is the square of a prime number.

The correct answer is B.

•

A researcher plans to identify each participant in a certain medical experiment with a code consisting of either a single letter or a pair of distinct letters written in alphabetical order. What is the least number of letters that can be used if there are 12 participants, and each participant is to receive a different code?

A. 4

B. 5

C. 6

D. 7

E. 8

Arithmetic Elementary Combinatorics

None of the essential aspects of the problem is affected if the letters are restricted to be the first n letters of the alphabet, for various positive integers n. With the 3 letters a, b, and c, there are 6 codes: a, b, c, ab, ac, and bc. With the 4 letters a, b, c, and d, there are 10 codes: a, b, c, d, ab, ac, ad, bc, bd and cd. Clearly, more than 12 codes are possible with 5 or more letters, so the least number of letters that can be used is 5.

The correct answer is B.

•

An object thrown directly upward is at a height of h feet after t seconds, where $h = -16(t-3)^2 + 150$. At what height, in feet, is the object 2 seconds after it reaches its maximum height?

A. 6

B. 86

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B. 86

C. 134

D. 150

E. 166

Algebra Applied Problems

Since $(t-3)^2$ is positive when $t \neq 3$ and zero when $t = 3$, it follows that the *minimum* value of $(t-3)^2$ occurs when $t = 3$. Therefore, the *maximum* value of $-16(t-3)^2$, and also the maximum value of $-16(t-3)^2 + 150$, occurs when $t = 3$. Hence, the height 2 seconds after the maximum height is the value of h when $t = 5$, or $-16(5-3)^2 + 150 = 86$.

The correct answer is B.

•

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